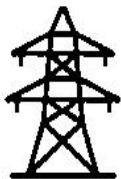


Advanced C&I Energy Solutions:

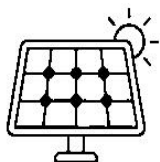
Integrating Solar, Storage, and Diesel Generation

Anhui UDAN Technology Co., Ltd.

The Hardware Foundation: Core System Components



Grid



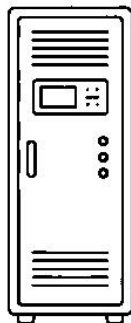
Solar PV



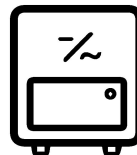
Diesel Generator



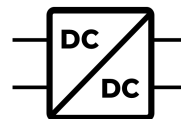
Battery Cluster



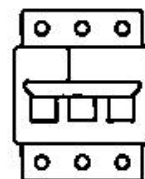
Energy Storage Cabinet



PCS



DC/DC Converter



AC Breaker



Meter



Loads

A Comprehensive Suite of Intelligent Control Strategies for Grid-tied System

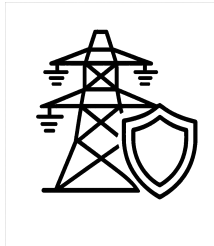
Economic Optimization

Strategies designed to reduce electricity costs and generate revenue.



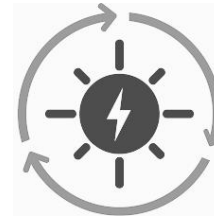
Grid Stability & Compliance

Strategies that ensure safe, reliable interaction with the utility grid.



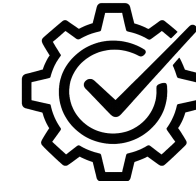
Resilience & Self-Consumption

Strategies focused on providing backup power, maximizing the use of on-site renewables.



System Integrity & Direct Control

Foundational layers for safety, protection, and manual oversight.

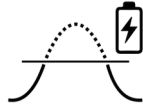


The EMS features 11 core strategies to optimize performance, ensure safety, and maximize financial returns. These capabilities are organized across four key functional pillars.

01

Peak Shaving & Load Shifting

- Arbitrages energy costs by charging during low-price periods and discharging during high-price(peak) periods.
- Default Flexible Control: Allows for the configuration of charging and discharging time windows, power ratings, and target State of Charge (SOC) levels.



02

Dynamic Capacity Firming

- Description: Controls the Energy Storage System (ESS) charge and discharge power to ensure that the Total System Load (ESS + Actual Facility Load) remains below a predefined threshold.
- Enable Switch: Toggle to enable or disable this function.
- Capacity Expansion Threshold: A user-defined setpoint. The EMS provides real-time control over ESS charging and discharging to ensure the grid meter reading stays below this value.

03

Demand Charge Management

- Description: Controls the Energy Storage System (ESS) charging power to ensure the Total System Load (ESS Charging + Facility Load) remains below the Demand Limit. (Current operation logic: The EMS tracks and records the Maximum Demand based on a 15-minute rolling window.)
- Enable Switch: Enable or disable the Demand Limiting function.
- Monthly Demand Reset (Initial Setpoint): Since the demand value is cumulative and tends to diverge (increase) as the load grows, the system requires a reset at the beginning of each month. The initial demand for the new month can be initialized as a percentage of the previous month's actual peak (e.g., Actual Peak $\times$ 70%, adjustable).

Strategies for Grid Stability & Compliance

01

Three-Phase Balancing

By enabling Automatic Three-Phase Power Regulation, the system dynamically adjusts the output power of each individual phase. This ensures stable system operation and effectively resolves issues related to Grid Phase Unbalance.



02

Overload Protection

- Description: Controls the Energy Storage System (ESS) charging power to ensure total power at the Point of Common Coupling (PCC/Transformer) does not exceed the maximum capacity.
- Enable Switch: Toggle to enable or disable this function.
- Charging Start Threshold: When the grid meter power is below the "Charging Start Threshold," the EMS allows the ESS to begin charging.
- Charging Limit Threshold: The EMS provides real-time control of the ESS charging power to ensure the total grid meter power remains below the "Charging Limit Threshold."
- Charging Inhibit Threshold: If load fluctuations exceed the EMS control capability, causing the grid meter power to rise above the "Charging Inhibit Threshold," the EMS will immediately set the ESS charging power to 0 to prevent overloading.

03

Anti-Export

Description: Controls the Energy Storage System (ESS) discharge power to prevent electricity from being fed back into the grid (backflow).

Enable Switch: Toggle to enable or disable the Anti-Backflow function.

Start Threshold: When the grid meter power exceeds the "Start Threshold," the EMS allows the ESS to begin discharging.

Discharge Limit Threshold: The EMS provides real-time control of the ESS discharge power to ensure the grid meter power remains above the "Discharge Limit Threshold" (maintaining a safety buffer from zero).

Inhibit Discharge Threshold: If load fluctuations exceed the EMS control capability, causing the grid meter power to fall below the "Inhibit Discharge Threshold," the EMS will command the ESS discharge power to 0 to prevent export.

Strategies for Resilience & PV Self-Consumption

01

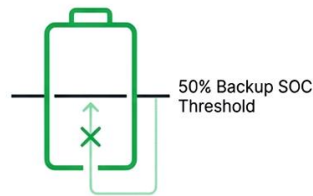
Backup Power Mode

Description: Ensures the battery maintains a minimum state of charge (SOC) to be ready for a grid outage.

Enable Switch: This function can be enabled or disabled.

Backup SOC Threshold : When SOC falls below this value (e.g., 50%), the EMS initiates charging from the grid or PV.

Release SOC Threshold : After backup is activated, when ESS SOC > "release threshold", EMS stops charging.



02

PV Strategy

Description: Prioritizes the use of solar energy to maximize savings and energy independence.

PV Configuration Scenarios:

① AC-Coupled PV: The PV system is connected to the AC side of the PCS .

② DC-Coupled PV: The PV system is connected to the DC side of the PCS.

PV Priority Strategy: Solar generation first serves the immediate facility loads excess solar energy is used to charge the battery.

PV priority – enable charging power: When grid meter power < threshold, EMS allows ESS charging.

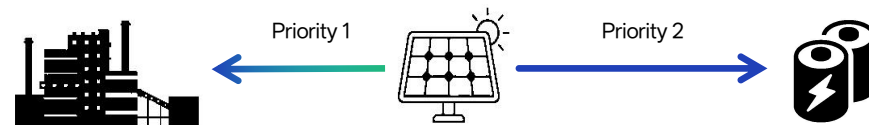
PV priority – disable charging power: When grid meter power > threshold, EMS stops ESS charging.

PV power verification: If PV meter exists or PV is directly connected to EMS, enabling this improves control accuracy.

PV maximum power setting: EMS limits PV generation so it does not exceed this value.

PV anti-export enable: EMS controls whether PV is allowed to feed power into the grid.

PV anti-export threshold: EMS adjusts PV output so grid meter power remains above this threshold.



Strategies for System Integrity & Direct Control

01

Secondary Protection (Level 2 Protection)

- SOC Upper/Lower Limits: Charging stops when SOC reaches this value.
- Charge/Discharge cut-off voltage: Charge/Discharge stops when the minimum cell voltage reaches this value.
- Release charge/discharge voltage: After charge/discharge cut-off, charge/discharge is allowed again once maximum/minimum cell voltage rises above this value.

02

Manual Control

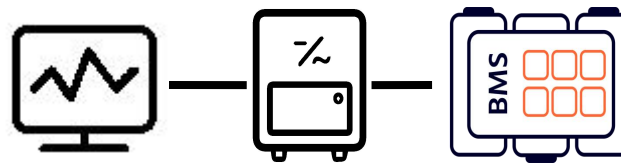
- In manual mode, users can manually control ESS power.
- Automatic: The ESS returns to automatic mode and follows peak-shaving strategies.
- Total power: Manually set charge/discharge power for the ESS.
- Subsystems: When EMS controls multiple PCS units, individual PCS power can be set manually.



03

Equipment Control

Enables direct operational commands for connected hardware, including startup, shutdown, and fault resets for BMS, PCS, and DC/DC units, units, as well as parameter control for cooling systems.



Diesel Generator (DG) Control Strategies

01

Diesel Generator(DG) Startup Logic



Functional Description: When the Diesel Generator is under EMS control, this section defines the specific conditions required to trigger the DG startup or shutdown.

1. Battery Voltage Thresholds

Startup Voltage: When the minimum battery cell voltage falls below this threshold, the EMS sends a command to start the DG.

Shutdown Voltage: When the minimum battery cell voltage rises above this threshold, the EMS sends a command to shut down the DG.

2. Battery SOC Thresholds

Startup SOC: When the battery SOC falls below this threshold, the EMS sends a command to start the DG.

Shutdown SOC: When the battery SOC rises above this threshold, the EMS sends a command to shut down the DG.

3. DO Configuration (Digital Output)

Users can bind a specific Digital Output (DO) port to physically control the start/stop signal of the Diesel Generator.

02

Gen-set Efficiency Optimization



Functional Description: Once the Diesel Generator is synchronized with the grid, the Energy Storage System (ESS) dynamically controls charging and discharging to maintain the DG load rate at a target of approximately 80%. In cases of significant load fluctuations, the target load rate should be adjusted downward. To achieve this, upper and lower limits for the target load rate must be configured.

Configuration Parameters:

A. Load Rate Thresholds

- Upper Load Limit: (Recommended maximum: 90%)
- Lower Load Limit: (Recommended minimum: 30%)
- If the actual load rate falls below the Lower Limit, the EMS will command the ESS to increase charging (or decrease discharging) to raise the DG's load.
- If the actual load rate exceeds the Upper Limit, the EMS will command the ESS to increase discharging (or decrease charging) to reduce the DG's load.

B. Adjustment Step Size

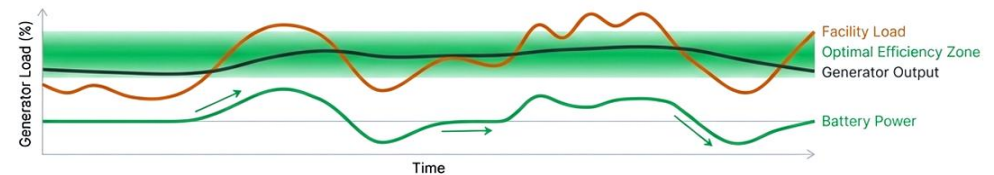
- The power adjustment rate for the ESS is configurable.
- Default Value: 10 kW/s (User-adjustable).

C. Function Enable/Disable

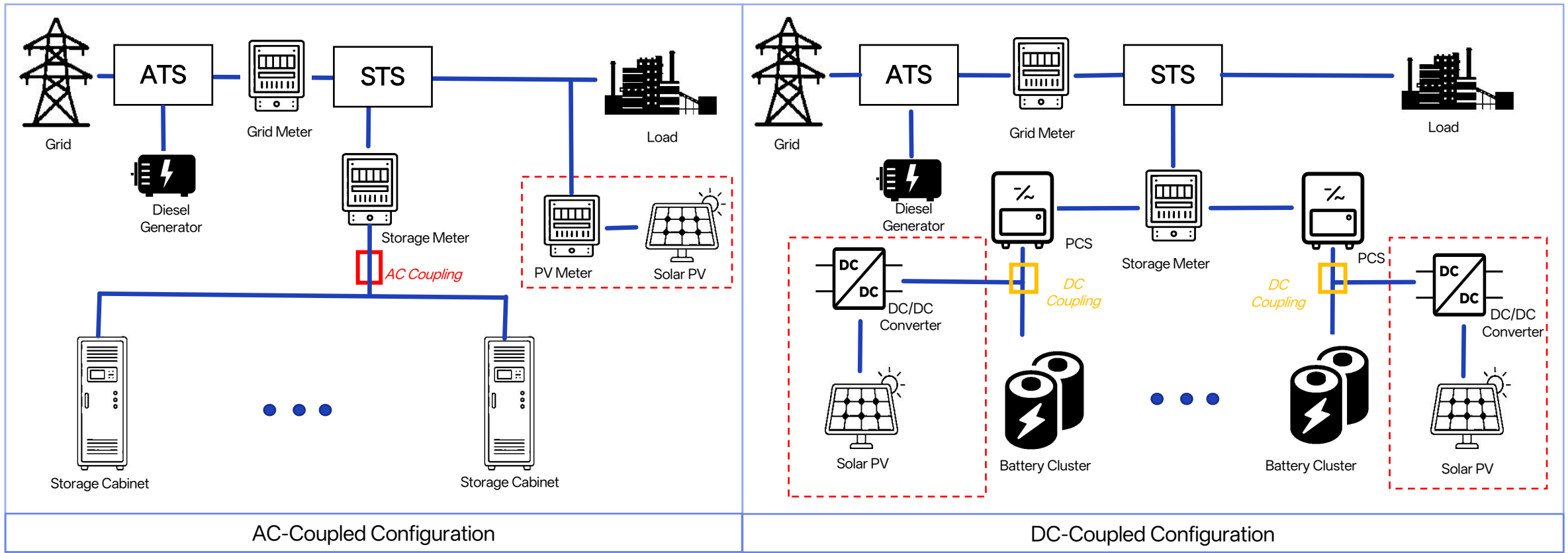
- A master switch to toggle this specific load-tracking feature ON or OFF.

D. DG Rated Power

- Enter the Nominal/Rated Power value of the Diesel Generator (used as the baseline for load rate calculations).



STS + ATS Coordinated Control: Transition between DG & Grid



1. Grid-Connected / Off-Grid Switching Strategy

- The grid-connected / off-grid mode is directly controlled by the coordination between the STS and the PCS.
- The EMS follows the grid-connected / off-grid status feedback from the PCS and switches to the corresponding control mode.
- Grid-connected mode: EMS controls power based on the grid-connected scheduled power.
- Off-grid mode: EMS cannot control PCS power directly. Only when the battery is fully charged or over-discharged (with off-grid parameters also checked), the EMS will Stop PCS, or Stop PV, or Limit current.

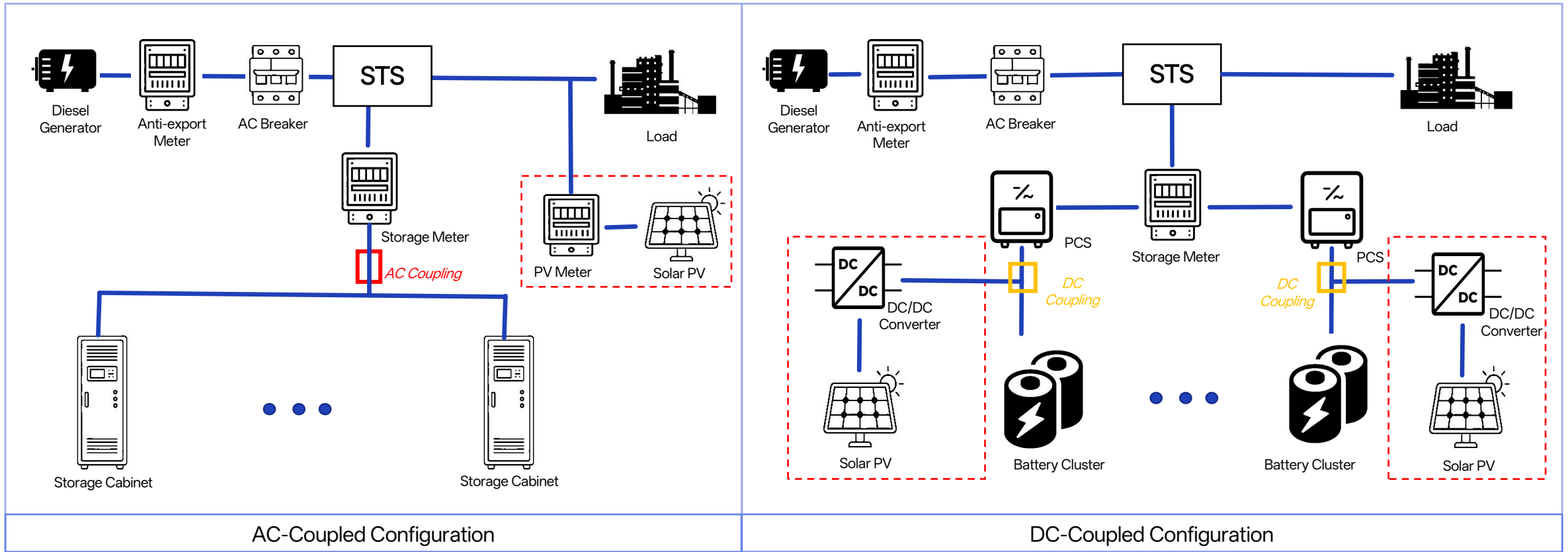
2. ATS Switching Strategy

- The ATS automatically determines the status of the utility grid and the diesel generator and switches the power source accordingly.
- The ATS informs the EMS of the current power source via DI signals.

3. Notes

- The PV inverter has strict requirements on grid frequency for startup.
- After the diesel generator starts, the PV inverter may fail to start.
- Therefore, PV may not be able to charge the energy storage system.
- The energy storage system can be charged with low power (e.g., 5 kW).
- If the PV system can operate:
 - The PV generation power should be limited to keep the diesel generator operating efficiency above 80% (configurable).
 - At the same time, enable ESS charging power to maximize PV energy utilization.

AC Breaker Coordinated Control: DG Seamless Transfer without Grid



1. Grid-Connected / Off-Grid Switching Strategy

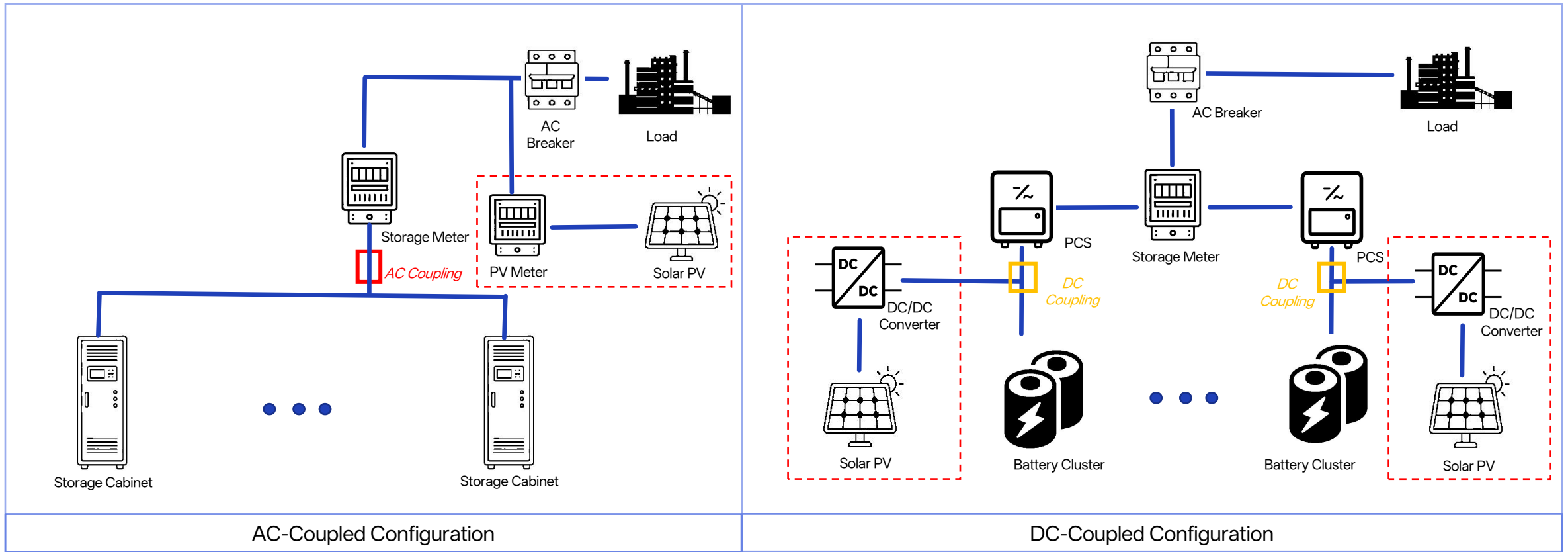
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- Grid-connected mode: EMS controls power based on the grid-connected scheduled power.
- Off-grid mode: EMS cannot control PCS power directly. Only when the battery is fully charged or over-discharged (with off-grid parameters also checked), the EMS will Stop PCS, or Stop PV, or Limit current.

2. AC Circuit Breaker Switching Strategy

- (1) Before the diesel generator (DG) starts, EMS defaults to controlling the AC circuit breaker to open.
- (2) After EMS controls the DG startup, and the DG's frequency and voltage are stable for 5 seconds:
 - A. Control the AC circuit breaker to close.
 - B. STS automatically switches to grid-connected/off-grid mode.
 - C. If energy storage and PV need to be connected to charge the system, EMS will issue power control to PCS and PV.
- (3) When the battery state of charge reaches the threshold, DG shutdown control is required:
 - First, control the DG to shut down.
 - Wait for STS to automatically switch to off-grid mode and remain stable.
 - Disconnect the AC circuit breaker.

****In this solution, the AC circuit breaker ensures that the energy storage system only engages after the diesel generator stabilizes, preventing frequent switching of STS during DG startup.**

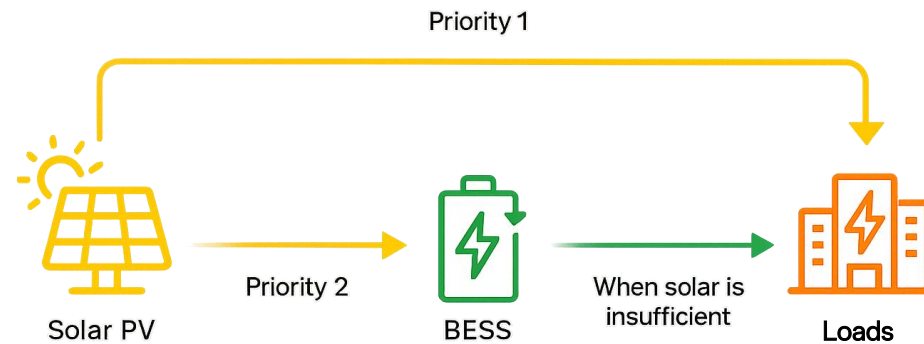
Dedicated Configuration: Pure Off-Grid Strategy



Strategies Descriptions

- This configuration is designed for full energy independence.
- The EMS prioritizes powering the load from available solar generation.
- Excess solar power is used to charge the battery.
- When solar power is insufficient, the battery discharges to meet the load demand.

Limitation: *In the absence of PV generation for a long time, the battery may be over-discharged and cannot be recharged.*



THANK YOU